

SageMaker Built-In Algorithms

Ready-to-Use Machine Learning, Without the Training Code

AWS Machine Learning Associate • Part 15

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What We'll Cover



What built-in algorithms are



How you use them



Supervised & unsupervised algorithms



Text, forecasting & vision



Choosing the right one



Built-in vs. your own code

What Are Built-In Algorithms?



Ready-made algorithms

- **Implemented & optimized by AWS**
  Packaged as containers, so no training code to write
- **Built to scale**
  Many support distributed training on large datasets
- **Simple to run**
  Data in S3, pick hyperparameters, train

A great starting point when your problem matches a common pattern, like predicting from tabular data, clustering, or forecasting.

How You Use Them

1



Pick the Algorithm

Match it to your problem type

2



Prepare Data

CSV or RecordIO-protobuf, stored in S3

3



Configure

Hyperparameters, instance type & count

4



Train & Deploy

Then host the model on an endpoint

Each algorithm lists its supported data formats and instance types in the docs.

The Categories at a Glance



Supervised

Learns from labels

XGBoost, Linear Learner, k-NN



Unsupervised

Finds structure

K-Means, PCA, Random Cut Forest



Text

Language tasks

BlazingText, Seq2Seq, LDA



Forecasting

Time series

DeepAR



Vision

Images

Classification, detection, segmentation

Supervised Algorithms



Linear Learner

Linear models for regression & classification. Simple, fast, a good baseline



XGBoost

Gradient boosted trees. A top choice for tabular regression & classification



K-Nearest Neighbors

Predicts from the closest examples in your data

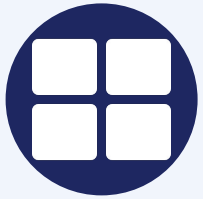


Factorization Machines

Built for sparse data, like recommendations & click prediction

These learn from labeled data.

Unsupervised Algorithms



K-Means

Groups similar points into clusters, like customer segments



PCA

Reduces the number of features while keeping most of the information



Random Cut Forest

Detects anomalies, like unusual sensor spikes or fraud



LDA & Neural Topic Model

Discover topics in a collection of documents

Also: IP Insights flags suspicious behavior around IP addresses.

Text, Forecasting & Vision



BlazingText

Word embeddings & fast text classification



Seq2Seq

Turns one sequence into another: translation, summarization



DeepAR

Forecasting that learns from many related time series at once



Image Algorithms

Image classification, object detection & semantic segmentation

Choosing the Right Algorithm

Predict a number or category from a table	**XGBoost / Linear Learner**
Group similar items, no labels	**K-Means**
Too many features to work with	**PCA**
Spot unusual data points	**Random Cut Forest**
Forecast future values of a time series	**DeepAR**

Practical Tips



Automatic Model Tuning

Works with the built-in algorithms to search for the best hyperparameters



Data Formats

Many work best with RecordIO-protobuf; pipe mode can stream it from S3



Instance Types

Image algorithms need GPUs; many tabular algorithms run fine on CPU



Data Quality Still Matters

Cleaning, encoding & scaling still apply

Built-In vs. Your Own Code

A standard problem, and you want to start quickly	**Built-In Algorithm**
A custom architecture or special logic	**Your Own Script or Container**
Start from a pretrained or foundation model	**SageMaker JumpStart**

Recap

- Built-in algorithms are optimized and ready to use, with no training code.
- Supervised, unsupervised, text, forecasting & vision cover many common problems.
- Match the problem to the algorithm, check the data format, and tune.

Thanks for watching!